CMS Open Payments Data Analyst

A natural-language interface to the **CMS Open Payments** program — the national disclosure dataset of financial relationships between pharmaceutical and medical device manufacturers and U.S. physicians and teaching hospitals, covering **calendar years 2018 through 2024**.

Currently loaded in this instance:

- **82,290,796** general payment records
- **5,936,201** research payment records
- **27,480** ownership interest records
- **18,932** removed / deleted records

*Counts are queried live against the local DuckDB database at startup — **88,273,409** total rows.*

Ask questions in plain English. The agent translates them to DuckDB SQL, runs them locally, and explains the results. Everything runs on your machine — no cloud, no API keys, no data leaves the host.

Example questions

- *Top 10 companies by total payment amount across all years*
- *Which medical specialties received the most general payments in 2024?*
- *Top 5 therapeutic areas by research funding in 2024*
- *Monthly payment trends for 2024*
- *How many physicians have ownership interests across all years?*

How it works

Each turn, the agent generates a DuckDB SQL query from your question (visible as a collapsible "Generating SQL" step), executes it against the local database, and summarizes the result in plain English. If the first SQL has an error, it self-corrects and retries up to 3 times.

Stack

- **DuckDB** for analytical SQL over Parquet files
- **Ollama** + **Qwen2.5-Coder 14B** for local Text-to-SQL
- **Chainlit** for the chat UI
- **Plotly** for interactive charts

See the project README and `phase-*.md` plans on disk for the full design.